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Chemoenzymatically Prepared Heparan Sulfate Containing Rare 2-O-Sulfonated Glucuronic Acid Residues

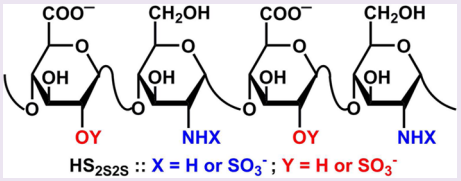
Rio S. Boothello,[†,#] Aurijit Sarkar,[†,#] Vy My Tran,[‡,#] Thao Kim Nu Nguyen,[‡] Nehru Viji Sankaranarayanan,[†] Akul Y. Mehta,[†] AlHumaidi Alabbas,[†] Spencer Brown,[‡] Alessandro Rossi,[‡] April C. Joice,[‡] Caitlin P. Mencio,[§] Maritza V. Quintero,[‡] Balagurunathan Kuberan,[‡,§,∥] and Umesh R. Desai*[,†]

[†]Institute for Structural Biology & Drug Discovery and Department of Medicinal Chemistry, Virginia Commonwealth University, Richmond, Virginia 23298, United States

[‡]Department of Medicinal Chemistry, [§]Interdepartmental Program in Neuroscience, and [∥]Department of Bioengineering, University of Utah, Salt Lake City, Utah 84112, United States

Supporting Information

ABSTRACT: The structural diversity of natural sulfated glycosaminoglycans (GAGs) presents major promise for discovery of chemical biology tools or therapeutic agents. Yet, few GAGs have been identified so far to exhibit this promise. We reasoned that a simple approach to identify such GAGs is to explore sequences containing rare residues, for example, 2-O-sulfonated glucuronic acid (GlcA*p*2S). Genetic algorithm-based computational docking and filtering suggested that GlcA*p*2S containing heparan sulfate (HS) may exhibit highly selective recognition of antithrombin, a key plasma clot regulator. HS containing only GlcA*p*2S and 2-N-sulfonated glucosamine residues, labeled as HS_{2S2S}, was chemoenzymatically synthesized in just two steps and was found to preferentially bind antithrombin over heparin cofactor II, a closely related serpin. Likewise, HS_{2S2S} directly inhibited thrombin but not factor Xa, a closely related protease. The results show that a HS containing rare GlcA*p*2S residues exhibits the unusual property of selective antithrombin activation and direct thrombin inhibition. More importantly, HS_{2S2S} is also the first molecule to activate antithrombin nearly as well as the heparin pentasaccharide although being completely devoid of the critical 3-*O*-sulfonate group. Thus, this work shows that novel functions and mechanisms may be uncovered by studying rare GAG residues/sequences.



Sulfated glycosaminoglycans (GAGs), natural linear copolymers of hexuronic acid and hexosamine residues, represent nature's bounty of chemical biology tools and therapeutic agents. Despite this, the current state-of-art displays minimal exploitation of this promise. Few GAG-based agents have been used to elucidate new pathways and very few GAG-based agents, other than polymeric heparin and dermatan sulfate, have reached the clinic. A major challenge in realizing the true biochemical and pharmacologic potential of GAGs is the structural microheterogeneity present in the GAGs isolated from tissues or cells. The variations introduced by non-template-driven sulfation, deacetylation, and epimerization reactions during their biosynthesis,[1,2] coupled with the postsynthetic fine-tuning of their structure by sulfatases,[3] generate millions of distinct sequences that pose significant challenges for detailed structural analysis.

Despite these challenges, sulfated GAGs are very attractive as potential seeds for new drugs. The presence of *O*-sulfonate (—OSO_3^-) groups on a GAG chain induces recognition and modulation of most proteins, especially those that display a collection of arginine and lysine residues on their surface. This gives rise to GAG modulation of a large number of physiological and pathological responses such as growth and cancer, cell renewal and differentiation, hemostasis and fibrinolysis, inflammation and immune response, and microbial invasion and defense.[2] It is likely that individual proteins bind to only a subset of GAG sequences from the natural repertoire of millions. A classic example of this high selectivity is the heparin pentasaccharide sequence DEFGH, containing a rare 3-O-sulfonated glucosamine (GlcN*p*) residue F, which binds to antithrombin (AT), a serpin involved in hemostatic regulation, with an affinity of ~50 nM.[4] Another example is a dermatan sulfate hexasaccharide, which binds heparin cofactor II (HCII), a serpin present in human plasma, with high selectivity and an affinity of 20 μM.[5] Although AT and HCII are related to each other in terms of three-dimensional structural similarity; they prefer completely different GAG sequences.[6] Such selective relationships can also be expected between other GAG sequence(s) and protein target(s).

How can such distinct GAG sequences be identified as promising chemical biology or drug discovery tools? An oft utilized approach in this process is the computational study of individual GAG–protein interactions.[7] This has yielded the prediction of GAG binding sites on enzymes,[8,9] enzyme

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inhibitors,[6,10,11] and other proteins.[12,13] Likewise, interesting GAG sequences have been also identified through the use of microarrays[14−16] and library synthesis.[17−19] Yet, whether GAG sequences identified from such approaches exhibit high selectivity for target proteins remains unclear. In fact, few GAG sequences have been identified that display high selectivity.

We reasoned that a simple approach to identify GAG sequences is to utilize nature's cues. GAGs present a small number of residues that are relatively rarely populated in natural preparations. We posited that such residues/sequences are likely to be biosynthesized by nature so as to elicit a selective or specific response. A classic example is the 3-*O*-sulfonated GlcN*p* residue, which is critical for selective modulation of AT.[4,10] This residue is also part of the heparin octasaccharide sequence that binds glycoprotein D of the herpes simplex virus and cyclophilin B, an inflammatory mediator.[20,21] No other rare GAG residue has been discovered to date as equally interesting from the chemical biology perspective.

We focused on the 2-*O*-sulfonated glucuronic acid (GlcA*p*2S) residue, which is a rare GAG residue without a well-established role. This residue can be biosynthesized by the action of 2-*O*-sulfotransferase (HS2ST) on GlcA*p*, which may be present in a growing heparan sulfate (HS) chain.[1,22,23] Under normal biosynthetic conditions, the growing HS chain is acted upon by *N*-deacetylase/*N*-sulfotransferase (NDST) to produce GlcN*p*2S and GlcA*p* containing sequences, a majority of which are further modified to IdoA*p* by C-5-epimerase (GLCE).[1,2] HS2ST then acts upon IdoA*p* to produce IdoA*p*2S residues, which give rise to the most common disaccharide sequence, IdoA*p*2S(1 → 4)GlcN*p*2S6S, in heparin. Yet, GLCE skips epimerization of a small number of GlcA*p* residues, which can be acted upon by HS2ST to form the rare GlcA*p*2S residues.[22,23]

This begged the question: What biological consequences would ensue if the GlcA*p*2S residue was more abundant in a GAG polymer? We used a genetic algorithm-based computational approach to study GlcA*p*2S containing hexasaccharides binding to AT and discovered elements of highly selective recognition. The HS polymer, labeled as HS_{2S2S}, and containing only GlcA*p*2S and GlcN*p*2S residues was prepared in a chemoenzymatic manner and found to potently bind and activate AT for inhibition of factor Xa. In addition, HS_{2S2S} was found to target HCII and thrombin, but not factor Xa. The results convey the potential of HS_{2S2S} in understanding differential recognition of proteins of the coagulation cascade, while also highlighting the significance of studying distinct GAG sequences for pharmacological purposes.

RESULTS

HS Sequences Containing GlcA*p*2S May Target Antithrombin. Although the GlcA*p*2S residue is rare in natural HS,[22,23] its proportion is enhanced 2−3-fold in the cerebral cortex compared with other tissues.[24] Only one isoform of HS2ST, the enzyme that gives GlcA*p*2S, has been identified to date, which suggests that the generation of GlcA*p*2S containing sequences is likely to be a finely tuned process.[22] Interestingly, HS3ST2, a 3-*O*-sulfotranferase isoform that introduces a rare 3-*O*-sulfonate group, appears to selectively recognize HS sequences containing GlcA*p*2S.[25] Further, this HS3ST isoform is expressed more in the brain,[26] which is also the place where GlcA*p*2S proportion is found to be higher.[24] This anecdotal evidence suggest that GlcA*p*2S is likely to be important.

Although HS3ST2 has not yet been linked to the biosynthesis of the high affinity heparin pentasaccharide sequence DEFGH, which specifically recognizes AT, we speculated on the possibility of the importance of high levels of GlcA*p*2S in brain for interacting with AT and, thereby, contributing to local hemostasis. To assess whether a HS chain containing only GlcA*p*2S and GlcN*p*2S residues, referred to here as HS_{2S2S}, could interact with AT, we utilized a computational workflow that helps sort "specific" and "non-specific" GAG interactions. The workflow involved a "steady state with no duplicates" genetic algorithm (GA) to place HS_{2S2S} sequence containing hexasaccharides onto the heparin-binding site (HBS) of AT. A GA is an artificial intelligence program that explores the three-dimensional space around a binding site and attempts to find optimal position(s) for a ligand through an iterative process of natural selection, alternately understood as "survival-of-the-fittest" rules. For studying the library of HS_{2S2S} hexasaccharides binding to AT, we utilized GOLD[6,10] as the GA and assessed the tendency of GlcA*p*2S and GlcN*p*2S monosaccharides to localize to specific locales in the HBS of AT. The tendencies were quantified using a grid-based approach (see Supplementary Figure S1, Supporting Information). Interestingly, both monosaccharides displayed statistically significant enrichment ($p < 2.2e^{-16}$) in the locales that compare favorably with monosaccharides D (GlcN*p*2S6S) and E (GlcA*p*) residues of the heparin pentasaccharide sequence DEFGH known to bind AT with high affinity (Figure 1). It may be noted that the D and E residues have previously been shown as predominantly responsible for both affinity and initial recognition with antithrombin.[4] Significant enrichment at these locales suggested a high probability of favorable and selective recognition of AT.

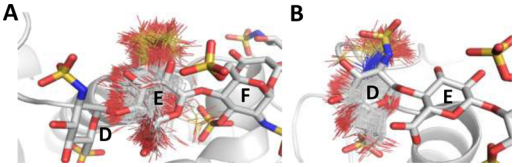


Figure 1. Genetic algorithm-based computational studies suggest a high probability of GlcA*p*2S and GlcN*p*2S recognizing antithrombin (AT) in a consistent manner. A library of 9918 HS hexasaccharide sequences, prepared from 38 disaccharide building blocks and containing at least one GlcA*p*2S and GlcN*p*2S residue, was docked onto the heparin-binding site of AT using GOLD,[51] following a reported protocol.[6,10] Analysis of clusters of poses obtained in the docking exercise (see Supplementary Figure S1, Supporting Information) revealed that GlcA*p*2S (A) and GlcN*p*2S (B) were found to bind repeatedly with RMSDs less than 1.2 Å suggesting a high specificity interaction.

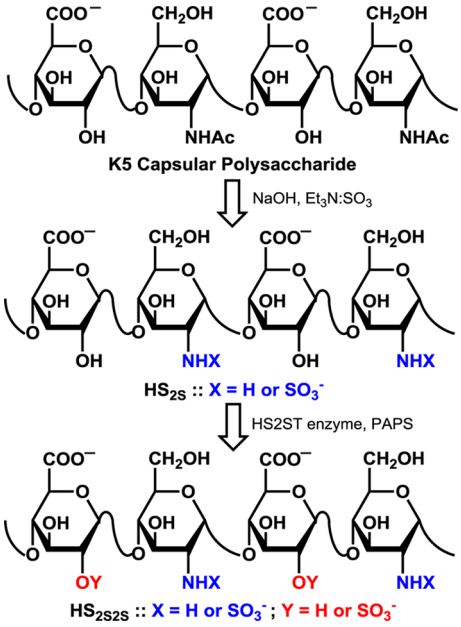
To evaluate the predicted interactions more quantitatively, the poses of each monosaccharide were parsed into two groups using k-means clustering to differentiate between specifically interacting and randomly interacting poses, as routinely performed for such studies.[27,28] The average pairwise RMSD for each group was then calculated. Interestingly, GlcA*p*2S and GlcN*p*2S were found to exhibit low RMSDs of 1.2 and 1.0 Å, respectively (see Supplementary Table S1, Supporting Information), which suggest a strong possibility of selective interaction with AT (Figure 1). It is important to note that a

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RMSD cutoff of 2.5 Å is typically used to differentiate "specific" from "nonspecific" interactions.[6,10] Thus, a HS chain containing GlcA*p*2S and GlcN*p*2S residues, that is, HS_{2S2S}, is computationally predicted to exhibit specific interaction with AT.

Chemoenzymatic Preparation and Characterization of HS_{2S2S}. Several efforts in the recent past have demonstrated the successful use of HS biosynthetic enzymes to produce targeted HS structures.[29−31] The availability of recombinant enzymes affords a robust avenue for preparing many different HS sequences. A special advantage of the HS_{2S2S} sequence is that it requires the use of only one enzyme, that is, HS2ST, to transfer sulfate groups onto the GlcA*p* residues of *N*-sulfoheparosan polymer, which can be prepared from the *Escherichia coli* K5 capsular polysaccharide (*N*-acetylheparosan) by the well-established chemical process of N-deacetylation followed by N-sulfonation.[32,33] Thus, the two step procedure affords HS_{2S2S} in reasonably good overall yields (~40%) starting from *N*-acetylheparosan (Scheme 1).

Scheme 1. Two-Step Synthesis of HS_{2S2S} Using a Chemoenzymatic Approach[a]



[a]The first step involves N-deacetylation and N-sulfonation of K5 capsular polysaccharide, *N*-acetylheparosan, followed by sulfation of the GlcA*p* using HS2ST.

Two simultaneous methods were utilized to ascertain the presence of GlcA*p*2S(1 → 4)GlcN*p*2S sequence in HS_{2S2S}. The digestion of the chemoenzymatically prepared polysaccharide using a mixture of heparin lyases I, II, and III was followed as a function of time through RP-IP UPLC methods optimized for resolving either disaccharides or oligosaccharides.[34−36] Interestingly, HS_{2S2S} appeared to exhibit considerable resistance to heparin lyase digestion in comparison to unfractionated heparin (UFH) suggesting considerable structural differences between the two related polymers. No cleavage of HS_{2S2S} was detected at 5 h in comparison to several disaccharides and oligosaccharides noted for UFH. Several tetrasaccharide peaks including T_{2S}, T_{3S}, and T_{4S} could be observed at 24 h for HS_{2S2S} (Figures 2A, see Supplementary Figure S3, Supporting Information, for individual structures), which eventually gave disaccharides D_{1S} and D_{2S} by 48 h (Figure 2B). The ESI-MS

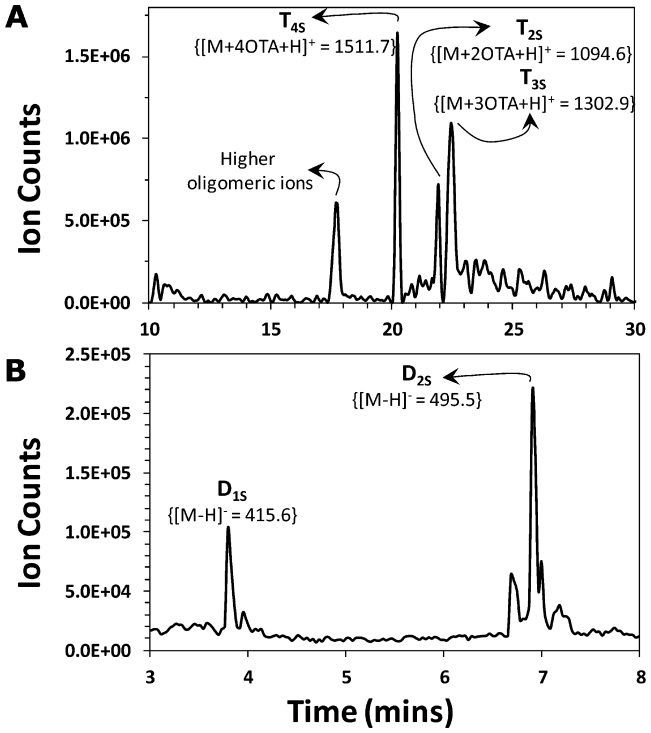


Figure 2. (A) Reversed phase-ion pairing (RP-IP) UPLC-ESI-MS total ion chromatogram obtained after 24 h digestion of HS_{2S2S} using a mixture of heparin lyases I, II, and III. The chromatogram shows the presence of tetrasaccharides with two, three, and four sulfate groups (T_{2S}, T_{3S}, and T_{4S}, respectively), while also showing undigested higher oligomeric ions. (B) RP-IP UPLC-ESI-MS chromatography of HS_{2S2S} following 48 h of digestion with heparin lyases I, II, and III. The chromatogram shows the presence of disaccharides with one and two sulfate groups (D_{1S} and D_{2S}, respectively). The gradients and counterions used in each chromatography were optimized for resolution of oligosaccharides (A) and disaccharides (B) based on literature reports. Experiments were performed in triplicate, of which one run is shown. OTA = octylamine counterion. See Materials and Methods for details.

profiles indicated that T_{2S}, T_{3S}, and T_{4S} corresponded to tetrasaccharides with two, three, and four sulfate/sulfamate groups, respectively (see Supplementary Figure S2, Supporting Information), while D_{1S} and D_{2S} corresponded to disaccharides with one and two sulfate/sulfamate groups. Furthermore, based on the elution times, the D_{1S} and D_{2S} correlate well with two heparin disaccharide standards suggesting the presence of GlcA*p*(1 → 4)Glc*p*N2S and GlcA*p*2S(1 → 4)Glc*p*N2S sequences in parent HS_{2S2S}, as expected. D_{2S} was found to be the major peak following nearly complete digestion. This indicated that HS_{2S2S} is composed of nearly 65% ± 7% GlcA*p*2S(1 → 4)Glc*p*N2S sequences (Figure 2B).

HS_{2S2S} Binds to AT with High Affinity. To put the computational prediction of HS_{2S2S}−AT interaction to test, we resorted to intrinsic tryptophan-based fluorescence spectroscopy, as reported in the literature.[4,37] A characteristic decrease followed by gradual saturation of the fluorescence emission at 340 nm (λ_{EX} = 280 nm) was observed under pH 7.0, *I* 0.06, 25 °C conditions, which could be analyzed using the standard quadratic binding eq 1 to yield an AT binding affinity of ~90 nM (Figure 3A, Table 1). In comparison, the AT affinities for full length UFH and heparin pentasaccharide DEFGH at pH 7.0, *I* 0.06, 25 °C were measured to be 20 and 43 nM, respectively (Table 1).[4,38] This implies that HS_{2S2S} appears to

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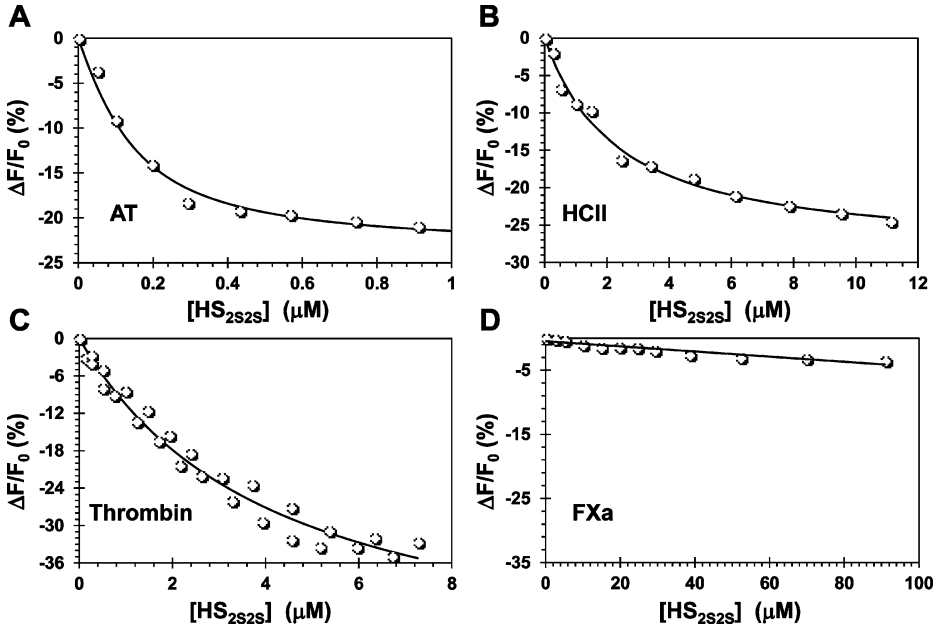


Figure 3. Representative titrations for the measurement of affinity of HS_{2S2S} for antithrombin (A), heparin cofactor II (B), thrombin (C), and FXa (factor Xa, D). Interaction of HS_{2S2S} with the proteins was studied by following the proportional change in fluorescence ($\Delta F/F_0$) as a function of concentration in 20 mM sodium phosphate buffer, pH 7.0, containing 25 mM NaCl at 25 °C. The saturable decrease in tryptophan (for AT, λ_{EX} = 280 nm, λ_{EM} = 340 nm), TNS (for HCII, λ_{EX} = 330 nm, λ_{EM} = 448 nm), dansyl (for FXa, λ_{EX} = 345 nm, λ_{EM} = 547 nm), or fluorescein (Thr, λ_{EX} = 490 nm, λ_{EM} = 520 nm) fluorescence was fitted using quadratic eq 1 (solid line) to derive the maximal fluorescence change (ΔF_{MAX}) and equilibrium dissociation constant (K_D). Essentially no change in fluorescein fluorescence was noted for FXa (D).

Table 1. Equilibrium Dissociation Constant (K_D) for GAG–Protein Complexes[a]

	K_D (μM)		
human protein	HS_{2S2S}	UFH	DEFGH
antithrombin	0.09 ± 0.03	0.02 ± 0.002	0.043 ± 0.007
heparin cofactor II	2.16 ± 0.27	0.416 ± 0.07	15 ± 3
thrombin	3.94 ± 0.56	0.54 ± 0.07	57 ± 7
factor Xa	≫100	30 ± 6	125 ± 35

[a]Measured using intrinsic (tryptophan) or extrinsic (fluoresceinylated protein) fluorescence in 20 mM sodium phosphate buffer, pH 7.0, containing 25 mM NaCl, 0.1 mM EDTA, 0.1% (w/v) PEG8000 at 25 °C. Error represents ± SD from at least two measurements. See Materials and Methods for additional details.

bind AT with slightly less affinity than UFH and DEFGH. However, considering the complete absence of the critical 3-O-sulfonated GlcN*p* residue in HS_{2S2S} (plus the absence of 6-O-sulfonated groups, which enhance the affinity of DEFGH),[4] its affinity for AT is high. In fact, for polymeric heparin devoid of the critical GlcN*p*2S3S6S residue, an AT affinity of 19 μM has been measured at pH 7.4, *I* 0.15, 25 °C,[39] which further supports the conclusion that HS_{2S2S} binds to AT with high affinity.

HS_{2S2S} Accelerates AT Inhibition of FXa More Than 200-fold. To assess the influence of HS_{2S2S} on the ability of AT to inhibit its two primary enzyme targets, factor Xa (FXa) and thrombin (TH), we measured the kinetics of enzyme inhibition in discontinuous assays at pH 7.0, *I* 0.06, 25 °C, as described earlier.[37−39] The exponential decrease in residual enzyme activity as a function of time was used to derive the observed pseudo-first order rate constant (k_{obs}) (Figure 4A,B). The profile of k_{obs} versus the concentration of HS_{2S2S}–AT complex was found to be linear (Figures 4D), as expected, which could be fitted by eq 2 to obtain the second-order rate constant for the uncatalyzed inhibition (k_{uncat}) from the intercept and HS_{2S2S} catalyzed inhibition of FXa from the slope (k_{HS}).

The k_{uncat} was found to be 1490 M^{-1} s^{-1}, which is lower than that measured at pH 7.4, *I* 0.15, 25 °C and in line with the expectations arising from the reduced activity of the serine protease at the lower pH of the experiment.[4,37] The k_{HS} was measured to be 323300 M^{-1} s^{-1} (Table 2), which translates to an increase of ∼217-fold in the rate of FXa inhibition in the presence of saturating levels of HS_{2S2S}. Similar results were observed in the presence of 2.5 mM $CaCl_2$. In comparison, heparin pentasaccharide DEFGH and full-length heparin UFH demonstrated accelerations of ∼270- and 834-fold, respectively (Table 2) under identical conditions. These results imply that HS_{2S2S} is weaker than the two reference GAGs by only ∼1.2- and ∼3.8-fold, respectively.[38,40] This implies that HS_{2S2S} is able to induce major acceleration in the efficacy of AT inhibition of FXa.

The k_{uncat} and k_{HS} for AT inhibition of TH in the presence of HS_{2S2S} were found to be 1060 and 80600 M^{-1} s^{-1}, respectively. This implies that at saturation HS_{2S2S} induces an acceleration of ∼76-fold in the rate of TH inhibition by AT (Table 2). More importantly, the absence of GlcN*p*2S3S6S in the full-length heparin chain does not introduce any defect in this acceleration.[39] DEFGH, on the other hand, induces only 1.2-fold increase in rate constant,[40] while UFH accelerates TH inhibition by AT nearly 3200-fold (Table 2).[38,40] These results indicate that HS_{2S2S} activation of AT for inhibiting TH appears to be different from that noted for both DEFGH and UFH.

Specificity of HS_{2S2S} Interaction with Coagulation Proteins. To assess whether HS_{2S2S} targets proteins closely related or relevant to the AT system, we studied its interaction with HCII, TH, and FXa using spectrofluorometry at pH 7.0, *I* 0.06, 25 °C. For these proteins, however, intrinsic tryptophan

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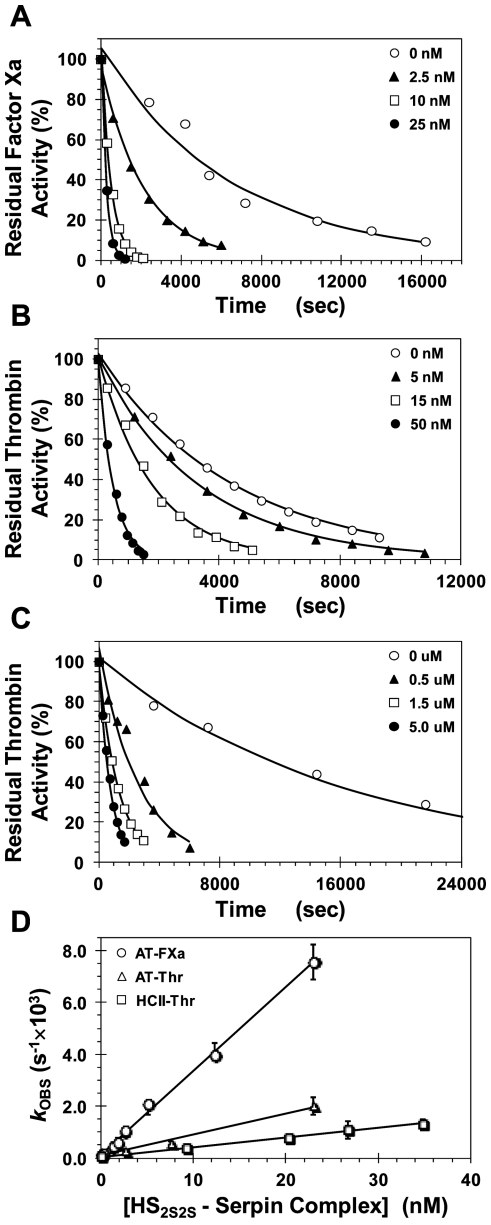


Figure 4. Kinetics of serpin (AT or HCII) inhibition of coagulation enzymes (Thr or FXa) in the presence of HS_{2S2S} in 20 mM sodium phosphate buffer, pH 7.0, containing 25 mM NaCl at 25 °C. The residual protease activity was measured from the initial rate of Spectrozyme FXa (A) or Spectrozyme TH (B, C) hydrolysis under pseudo-first-order conditions as a function of time in the presence of different concentrations of HS_{2S2S} and fixed concentrations of AT (A) and HCII (B, C). The profile of the observed pseudo-first-order rate constant of inhibition (k_{obs}) at each HS_{2S2S} concentration was analyzed using linear eq 2 to calculate the uncatalyzed (k_{uncat}) and catalyzed (k_{HS}) rate constant of inhibition from the intercept and slope of the fitted lines (solid lines).

fluorescence could not be used as effectively because the change in the target protein signal was too small. Hence, extrinsic fluorophores were utilized including TNS for HCII and fluorescein for TH and dansyl for FXa. Both extrinsic fluorophores have been utilized earlier for these proteins.[41,42] A saturable decrease in extrinsic fluorescence was observed for HS_{2S2S} binding to HCII as well as to TH, from which affinities of 2.2 and 3.9 μM, respectively, were calculated for the two interactions (Figure 3, Table 1). In contrast, essentially no change in fluorescence was observed for the HS_{2S2S}–FXa system suggesting minimal interaction ($K_D \gg 100$ μM). The results suggest that HS_{2S2S} binds to HCII and TH with fairly high affinity, although it prefers AT by at least 5.5-fold. Two systems that can be used for comparison here include the dermatan sulfate–HCII and UFH–TH systems, which have been reasonably well studied and display affinities in the range of 0.5–50 μM at pH 7.4.[43,44] Likewise, the affinity of a high affinity dermatan sulfate hexasaccharide for HCII has been measured to be 20 μM.[5] Thus, HS_{2S2S} binds to HCII and TH with affinities that either compare favorably to or are better than other well studied GAGs.

HS_{2S2S} Is Only a Marginal Potentiator of HCII Inhibition of TH. To assess whether the dual interaction of HS_{2S2S} is an aid or barrier to HCII inhibition of TH, we studied the kinetics of inhibition (Figure 4). The second-order rate constant of HCII inhibition of TH was measured in a manner similar to that for AT through discontinuous measurement of pseudo-first-order rate constants at varying HS_{2S2S} concentrations. HCII inhibition of TH was accelerated ~34-fold by HS_{2S2S} (Table 2). In comparison, UFH was found to accelerate this reaction some 2090-fold (Table 2), which is essentially identical to that reported for full-length dermatan sulfate.[43] The HCII–UFH system is considered to be a nonspecific system because no heparin sequence capable of selectively binding to HCII has been identified as yet. This implies that even if HS_{2S2S} was binding to HCII in a nonspecific manner, an activation significantly higher than 34-fold should have been observed. Thus, the results suggest that HS_{2S2S} accelerates HCII inhibition of TH only marginally.

HS_{2S2S} Directly Inhibits TH But Not FXa. To assess whether HS_{2S2S} directly inhibits TH and FXa, we measured their residual activity in the presence of HS_{2S2S} using chromogenic substrate hydrolysis assay, as described earlier.[44] Interestingly, TH activity decreased as a function of HS_{2S2S} concentration, which could be fitted using the logistic dose–response eq 3 to obtain an IC_{50} of 4.0 ± 0.3 μM (Figure 5) and an efficacy of 40% ± 1%. In contrast, FXa activity remained essentially unaffected even at 100 μM HS_{2S2S}. Although a large number of studies report GAG binding to TH, this is the first demonstration of direct TH inhibition by a natural or synthetic GAG. Our earlier studies on GAG recognition of TH have shown its highly surface exposed character arising from the involvement of highly flexible side chains,[45] which disfavors allosteric modulation of the active site located nearly 20 Å away. Thus, TH inhibition by HS_{2S2S} probably arises from GlcA*p*2S and/or GlcN*p*2S residues interacting with the protease in a manner considerably different from the parent highly sulfated GAG sequences.

DISCUSSION

This work shows that our original hypothesis on studying rare GAG residues to discover some selective chemical biology or therapeutic tools is promising and merits further investigation. In principle, selectivity can be of two types, chemical and biological. The former refers to the idea that one sequence out of many selectively recognizes the target, whereas the latter refers to the idea that one binding mode out of the many possible in a target site is selectively preferred. Computationally, the HS_{2S2S} sequence was deduced from among the many present in the library, which alludes to elements of chemical specificity. With regard to biological specificity, HS_{2S2S} was found to preferentially target AT in comparison to HCII

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Table 2. Acceleration in Serpin Inhibition of Coagulation Enzymes Brought about by HS_{2S2S}, UFH, and DEFGH[a]

serpin−enzyme system	k_{uncat} (M^{-1} s^{-1})	k_{HS}	acceleration[b] (HS_{2S2S})	acceleration[b] (UFH)	acceleration[b] (DEFGH)
AT−FXa	1490 ± 300[c]	323300 ± 7560	217 ± 34	848 ± 34	270 ± 18
AT−TH	1060 ± 200	80600 ± 8200	76 ± 15	3223 ± 87	1.2 ± 0.4
HCII−TH	1090 ± 200	36800 ± 1320	34 ± 4	2090 ± 40	45 ± 2

[a]Measured using discontinuous enzyme inhibition assay in 20 mM sodium phosphate buffer, pH 7.0, containing 25 mM NaCl, 0.1 mM EDTA, 0.1% (w/v) PEG8000 at 25 °C. See Materials and Methods for additional details. [b]Refers to the ratio of catalyzed and uncatalyzed rate constants. [c]Error represents ± SD from at least two measurements.

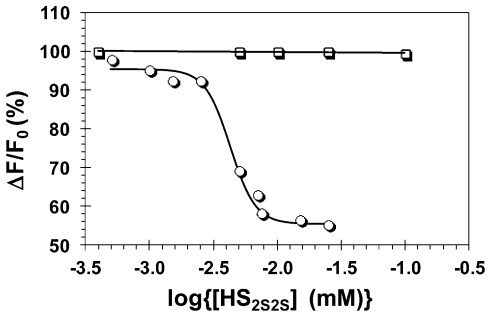


Figure 5. Studies on direct inhibition of thrombin (○) and factor Xa (□) by HS_{2S2S}. Protease inhibition was measured spectrophotometrically through chromogenic substrate hydrolysis assay in 20 mM sodium phosphate, pH 7.0, containing 25 mM NaCl at 25 °C. Solid line for thrombin represents nonlinear fit to the data using eq 3 to obtain IC_{50}, Y_M, and Y_0. Solid line for factor Xa represents linear regression.

suggesting selectivity of recognition between two closely related serpins. Likewise, direct inhibition of TH by HS_{2S2S} but not of FXa also conveys the same idea for two closely related enzymes. Thus, this work shows that HS chains containing 2-O-sulfonated glucuronic acid residues appear to display high level of selectivity predicted on the basis of first-principles involving rare residues.

With regard to distinct novel results, HS_{2S2S} accelerates AT inhibition of FXa more than 200-fold. This is a substantial level of acceleration considering that clinically used fondaparinux brings about a ∼300-fold effect.[4,40] Considering that HS_{2S2S} does not bind to FXa, the acceleration in enzyme inhibition is most probably brought about by induction of conformational changes in AT and not through the bridging mechanism.[4,40] This implies that the HS_{2S2S} sequence is another sequence in addition to the well-established heparin pentasaccharide sequence DEFGH that induces structural changes in the main body of the serpin, AT. Establishing the nature of these structural changes and the mode of interaction will be important to develop advanced sequences based on HS_{2S2S}, which is composed primarily of GlcA*p*2S(1 → 4)GlcN*p*2S sequence (>60%). Recent studies show that chemoenzymatically prepared HS hexasaccharides containing one GlcA*p*2S residue do not bind to AT.[46] Thus, we posit that the reason HS_{2S2S} binds and activates AT so well is because of the presence of several GlcA*p*2S residues within the serpin binding domain. The presence of several GlcA*p*2S units may enhance the probability of interaction with AT in a statistical manner or may enhance the selectivity of binding to AT because of a simultaneous engagement of two or more GlcA*p*2S residues. Although our initial search algorithm was not designed to seek such dual binding sequences, an analysis of the hexasaccharides whose GlcA*p*2S residues bind at the same location as E of DEFGH, as in Figure 1, often contained a second GlcA*p*2S residue at an adjacent site (not shown). This supports the possibility that multiple GlcA*p*2S residues, located not too far away from each other on the HS_{2S2S} chain, are important. A key conclusion of this observation is that a heparan sulfate oligosaccharide of appropriate length and composed of two or more GlcA*p*2S(1 → 4)GlcN*p*2S units may conformationally activate AT well (>200-fold) for use as an anticoagulant. Thus, HS_{2S2S} is the first molecule that activates antithrombin nearly as well as the heparin pentasaccharide although being completely devoid of the critical 3-*O*-sulfonate group.

HS_{2S2S} induces only a 76-fold increase in the rate of TH inhibition by AT, which is small in comparison to the ∼3000-fold brought about by clinically used UFH.[38,39] This is an unusual result. The key mechanism that brings about acceleration in AT inhibition of TH by UFH is the bridging mechanism, which is a two step process that organizes the UFH−AT−TH ternary complex. The first step involves specific interaction of the UFH with AT to form a binary UFH−AT complex. TH then binds nonspecifically to a remote site on the UFH chain and "walks" to the site of UFH−AT binary complex for accelerated inhibition.[38,47] Because the bridging mechanism is less dependent on the conformational change induced in AT by the GAG, most polymeric sulfated GAGs bring about significant acceleration in TH inhibition. Thus, *a priori* HS_{2S2S} should have induced high acceleration in AT inhibition of TH. The exact reason for the less-than-expected acceleration in AT inhibition of TH in the presence of HS_{2S2S} remains unclear at present because of the difficulties of studying the HS_{2S2S}−AT−TH template formation process. HS_{2S2S} by itself is an inhibitor of TH, which endangers the formation of the ternary complex because of the formation of inhibited HS_{2S2S}−TH binary complex ($K_D \approx 4$ μM). However, the results show that the template mediated AT inhibition of TH in the presence of HS_{2S2S} is less likely to be an important contributor.

Although defective in mediating AT acceleration of TH inhibition, HS_{2S2S} studies have led to the discovery of GAG-mediated direct inhibition of TH. To the best of our knowledge, HS_{2S2S} is the first polymeric GAG that demonstrates this property. Direct inhibition of thrombin is of considerable interest because of thrombin's role in hemostasis and angiogenesis.[48] In fact, we have previously reported several polymeric mimetics of sulfated GAGs as direct inhibitors of TH (and also of FXa).[49] These sulfated GAG mimetics were found to bind in exosite II of TH and induce catalytic dysfunction of the active site.[44] Likewise, a small highly sulfated molecule called sucrose octasulfate was also found to interact with exosite II and inhibit TH, albeit with an efficacy of only 5−10%.[50] Interestingly, HS_{2S2S} does not bind FXa, although its exosite II is similar (but not identical) to that of TH. Thus, either HS_{2S2S} inhibits thrombin by binding in a site completely different from exosite II, for example, exosite I, or is able to exploit subtle differences in the heparin-binding sites of two similar serine proteases. Further studies are necessary to clarify the molecular

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mechanism of this interesting process. Yet, the results allude to a novel idea that an oligosaccharide composed of two or more GlcA*p*2S(1 → 4)GlcN*p*2S units may directly target TH.

Finally, an important question to address is whether GlcA*p*2S containing HS is likely to possess a novel role *in vivo*. This work identifies a couple of biochemical processes where the novel sequence is shown to exhibit an anticoagulant function. However, it is important to recognize that although the GlcA*p*2S(1 → 4)GlcN*p*2S sequence is possible in nature,[1,22,23] the polymeric HS_{2S2S} sequence is an unnatural sequence. Considering that GAG biosynthesis is known to be regulated in a spatiotemporal manner,[1−3] the levels of GlcA*p*2S containing sequences (and its 3-*O*-sulfonate variants) may be elevated in selected tissues, for example, brain, at certain time points. Thus, studies with HS_{2S2S} may help in determining new molecular pathways. For example, a comprehensive study of the interactions of HS_{2S2S} with other GAG-binding serpins, for example, protease nexin 1, may further help to understand the role of GlcA*p*2S containing sequences. Likewise, it would be important to assess whether HS_{2S2S} is less likely to interact with platelet factor-4, the key mediator of heparin-induced thrombocytopenia. If it does not interact with platelet factor-4, then HS_{2S2S} or its analog sequence(s) may lead to new anticoagulants devoid of complications arising from heparin therapy.

Overall, we have put forward a novel approach of utilizing nature's cue regarding rare residue(s) to discover a chemoenzymatic HS that displays selective modulation of AT and TH function. These properties are not likely to be present in natural HS containing GlcA*p*2S residues because of their low natural abundance. HS_{2S2S} demonstrates novel function (AT activation without 3-O-sulfonated residue and direct TH inhibition), which are very useful, complementary, and amenable to further structural optimization. We believe that HS_{2S2S} may serve as a promising lead for the discovery of polysaccharides with dual action anticoagulant potential. More broadly speaking, the results highlight the significance of studying unique GAG sequences for use as chemical biology tools or for discovery of therapeutic agents.

MATERIALS AND METHODS

Proteins and Chemicals. Human plasma AT, HCII, TH, fluorescein-tagged TH (FFPRCK-TH), and fluorescein-tagged FXa (FXa-DEGR) were purchased from Haematologic Technologies (Essex Junction, VT) and used as such. The proteins were stored in 20 mM sodium phosphate buffer, pH 7.0, containing 25 mM NaCl, 0.1 mM EDTA, and 0.1% (w/v) PEG8000 at −80 °C. Spectrozyme FXa and thrombin were obtained from American Diagnostics. 2-(*p*-Toluidino)naphthalene-6-sulfonic acid (TNS) was purchased from Sigma-Aldrich. Anhydrous CH_3CN, NH_4OAc, and acetic acid were from Fisher Scientific. Tributylamine (TrBA) and octylamine (OTA) were purchased from Acros Organics. Heparin standards were obtained from Galen Laboratory Supplies, whereas the UFH was purchased from Sigma-Aldrich. The fast-flow DEAE-sepharose gel was purchased from GE Healthcare (Amersham Biosciences). The *E. coli* strain K5 (O10−K5 (L)-H4) was obtained from ATCC. 3′-Phosphoadenosine-5′-phosphosulfate was purchased from Sigma-Aldrich. All other reagents and solvents were high purity reagent grade.

Preparation of a Library of HS Hexasaccharides for Computational Studies. Thirty-eight HS disaccharide structures containing UA*p* and GlcN*p* were constructed using SybylX v. 2.0. The structures incorporated multiple ring conformations of all the known natural substitutions at the 2-, 3-, and 6-positions of the two residues. These disaccharides were then used to construct a combinatorial library of hexasaccharides starting with either a UA*p* or a GlcN*p* residue at the nonreducing end, which gave two libraries each containing 38 × 38 × 38 = 109744 unique H/HS hexasaccharide sequences. From these two libraries, sequences containing at least one →4)GlcA*p*2S*β*(1 → 4)GlcN*p*2S*α*(1→ or →4)GlcN*p*2S*α*(1 → 4)GlcA*p*2S*β*(1→ disaccharide were considered for further analysis. This led to 9918 unique such sequences.

Docking of HS_{2S2S}-Containing Hexasaccharide Library onto AT. The AT structure used in these studies was derived from Protein Databank entry 1TB6[47] and was prepared using Sybyl's "prepare protein" option. Hydrogens were added and minimized using the Tripos force field and Gasteiger−Hückel charges. The library of 9918 HS_{2S2S} sequence-containing hexasaccharides was docked into the heparin-binding site of AT using GOLD,[51] which uses a genetic algorithm to place ligands into the binding site. Each hexasaccharide structure was docked using 100 GA runs, each consisting of 100000 iterations. A 16 Å docking radius around the N^{ζ} atom of Lys125 was used, which is approximately in the center of the heparin-binding site of AT. The inter-glycosidic linkages of the hexasaccharides were maintained rigid at the average φ and ψ values derived from the literature.[10] The GA runs were allowed to terminate early if the top three solutions had an RMSD of 2.5 Å or lower, of which the top two poses were stored and analyzed at the end of the docking experiment. GoldScore was used to assess the fitness of the docked poses.

Grid-Based Screening of Protein−Monosaccharide Interactions. The overall scheme describing assessment of specific protein−monosaccharide interactions is described in Supplementary Figure S1, Supporting Information. In-house python scripts that incorporated the Openeye OEChem toolkit V 2.0.0 (OpenEye Scientific Software, Inc.) were used to investigate the residence tendencies of monosaccharides on protein HBSs. A grid was generated around the docked hexasaccharides to analyze preferences in the location of GlcA*p*2S and GlcN*p*2S residues. A density map was obtained by counting the number of times a monosaccharide resided closest to a grid point. Clustering of GlcA*p*2S and GlcN*p*2S residue poses was performed using the k-means clustering algorithm, as reported earlier.[52,53] The search algorithm focused on identifying two clusters with maximal distance between them. This enabled subsequent identification of the "tightest" cluster, that is, the one showing least deviation in atomic positions of the pyranose ring, as measured by the RMSD metric implemented in the Openeye OEChem toolkit. Statistical significance of enrichment of either GlcA*p*2S or GlcN*p*2S residue at its highest-density location in comparison to that at other locations on the grid was analyzed using a proportionality test in the R statistical programming environment.[55]

Synthesis of *N*-Sulfoheparosan. Sulfation of the heparosan polysaccharide was performed as described earlier.[32] Briefly, heparosan (1 g) in 100 mL of 2.5 M NaOH was stirred overnight at 55 °C, the solution was brought to pH 7.0, and 5 g each of Na_2CO_3 and triethylamine sulfotrioxide was added with stirring. The reaction was allowed to continue at 48 °C for 2 days following which the solution was centrifuged at 16000*g* for 5 min. The supernatant was collected, diluted with equal amount of deionized water, and loaded onto a 5 mL DEAE sepharose column, which was pre-equilibrated with 20 mM NaOAc buffer, pH 6.0, containing 0.1 M NaCl. Following washing (30 bed volumes), *N*-sulfoheparosan was eluted from the column using 20 mM NaOAc buffer, pH 6.0, containing 1 M NaCl. *N*-Sulfoheparosan was precipitated from the solution using ethanol and the lyophilized to obtain a white solid (~800 mg, ~80% yield).

HS2ST Modification of *N*-Sulfoheparosan. *N*-Sulfoheparosan (500 *μ*g) was incubated with PAPS (500 *μ*g) and HS2ST enzyme (~100 *μ*g) in 25 mM MES buffer, pH 7.0, containing 0.02% Triton X-100, 2.5 mM $MgCl_2$, 2.5 mM $MnCl_2$, 1.25 mM $CaCl_2$, and 0.75 mg mL^{-1} BSA at 37 °C for 72 h. The reaction mixture was purified through a mini DEAE-Sepharose column (0.2 mL) that had been pre-equilibrated with 20 mM NaOAc buffer, pH 6.0, containing 0.1 M NaCl and 0.01% Triton X-100. The bound HS_{2S2S} was eluted from the column using 20 mM NaOAc buffer, pH 6.0, containing 1 M NaCl. The HS_{2S2S} polysaccharide samples were desalted and concentrated with 3000 MWCO Amicon columns (~244 *μ*g, 49% yield).

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Enzymatic Digestion and Structure Elucidation of HS_{2S2S}. Cleavage of HS_{2S2S} (8 μg/μL) or porcine UFH (8 μg/μL) using a mixture of heparin lyases I, II, and III (5 mU each/mg substrate) was performed in deionized, high purity water, pH 7.0, at 37 °C. The mixture was directly injected into Waters Acquity H-Class UPLC coupled to a TQD ESI-MS detector every 5, 26, 48, and 75 h for analysis of smaller fragments formed upon elimination reaction. Two literature reported methods (with small modifications) were adopted to monitor the formation of disaccharides and oligosaccharides, following literature reports.[34−36] For oligosaccharides, reversed-phase separation of the digested mixture was achieved on an BEH C18 1.7 μm (2.1 mm × 150 mm) column at 40 °C using a linear gradient of 100% solvent A (water/CH_3CN (95:5, v/v)) to 100% solvent B (water/CH_3CN (15:85, v/v)) in 45 min at a flow rate 0.1 mL/min. Both solvents A and B contained 15 mM octylamine (OTA) as an ion-pairing reagent and 100 mM HFIP as organic modifier. The ESI-MS peaks were detected in positive ionization mode with a capillary voltage of 3.2 kV, extractor voltage of 1 V, cone voltage between 20 and 100 V, radio frequency of 0.1 V, source temperature of 150 °C, and desolvation temperature of 350 °C. To quantify disaccharides formed upon HS_{2S2S} digestion, BEH C18 1.7 μm (2.1 mm × 100 mm) column and a linear gradient of 100% solvent A (water/CH_3CN (95:5, v/v)) to 100% solvent B (water/CH_3CN (20:80, v/v)) were employed with 20 mM tributylamine (TrBA), 2.5 mM ammonium acetate and 22.2 mM acetic acid as ion-pairing agents, as previously reported.[34] The pH of the mobile phase was carefully adjusted at 7.0. The temperature of the column was maintained at 40 °C, while the flow rate was maintained at 0.5 mL/min. The ESI-MS detector was set to negative ionization mode using a capillary voltage of 3.0 kV, cone voltage between 20 and 40 V, extractor voltage of 1 V, radio frequency of 0.5 V, source temperature of 150 °C, and desolvation temperature of 350 °C.

Equilibrium Binding Studies Using Fluorescence Spectroscopy. The dissociation constant of HS_{2S2S}−protein and UFH− or DEFGH−protein complexes were measured using change in fluorescence emission as a function of the concentration of the GAG in 20 mM sodium phosphate buffer, pH 7.0, containing 25 mM NaCl, 0.1 mM EDTA, and 0.1% PEG8000 at 25 °C, as described earlier.[55] The experiments were performed using a QM4 fluorometer (Photon Technology International, Birmingham, NJ) in a quartz microcuvette by titrating it into a 200 μL solution of the protein (100−250 nM) and monitoring the change in the fluorescence at either 340 nm (for AT, λ_{EX} = 280 nm), 520 nm (for TH, λ_{EX} = 490 nm), 448 nm (for HCII, λ_{EX} = 330 nm), or 547 nm (for FXa, λ_{EX} = 345 nm). The concentration of TNS used for measuring affinity of HS_{2S2S} for HCII was 10 μM. Excitation and emission slit width were set to 1.0 mm. The saturable change in fluorescence signal was fitted using the quadratic equilibrium binding eq 1 to obtain the K_D of interaction. In this equation, ΔF represents the change in fluorescence at a fixed concentration of HS_{2S2S} from the initial fluorescence F_0 and ΔF_{MAX} represents the maximal change in fluorescence following saturation of the protein. $[P]_0$ represents the concentration of AT, HCII, TH, or FXa.

$$\frac{\Delta F}{F_0} = \frac{\Delta F_{max}}{[P]_0}\{(([P]_0 + [HS_{2S2S}]_0 + K_D) - \sqrt{(([P]_0 + [HS_{2S2S}]_0 + K_D)^2 - 4[P]_0[HS_{2S2S}]_0)})/2\} \quad (1)$$

Kinetics of Protease Inhibition in the Presence of HS_{2S2S}. The kinetics of inhibition of coagulation proteases, TH or FXa, by AT or HCII in the presence of HS_{2S2S} and UFH or DEFGH was measured spectrophotometrically using a microplate reader (FlexStation III, Molecular Devices) under pseudo-first-order conditions, as described earlier.[4,37] Briefly, a fixed concentration of TH or FXa (5 nM) was incubated with fixed concentrations of plasma AT (100 nM) or HCII (50 nM) and HS_{2S2S} (0−5000 nM) or UFH (0−2 nM)/DEFGH (0−900 nM) in 20 mM sodium phosphate buffer, pH 7.0, containing 25 mM NaCl, 0.1 mM EDTA, and 0.1% (w/v) PEG8000 at 25 °C. At regular time intervals, an aliquot of the inhibition reaction was quenched with 100 μL of 125−200 μM chromogenic substrate (Spectrozyme TH or Spectrozyme FXa) in 20 mM sodium phosphate buffer, pH 7.0, containing 25 mM NaCl at 25 °C. To determine the residual protease activity, the initial rate of substrate hydrolysis was measured from the increase in absorbance at 405 nm. The exponential decrease in the initial rate of substrate hydrolysis as a function of time was used to determine the observed pseudo-first-order rate constant of protease inhibition (k_{obs}). A plot of k_{obs} at different concentrations of HS_{2S2S}−serpin complex could be described by eq 2, in which k_{uncat} is the second-order rate constant of protease inhibition by serpin alone and k_{HS} is the second-order rate constant of protein inhibition by serpin−HS_{2S2S} complex (HS_{2S2S}/P).

$$k_{obs} = k_{uncat}[P]_0 + k_{HS}[HS_{2S2S}/P] \quad (2)$$

Direct Protease Inhibition in the Presence of HS_{2S2S}. Direct inhibition of TH or FXa was measured through a chromogenic substrate hydrolysis assay using a microplate reader (FlexStation III, Molecular Devices), as described earlier.[42,44] Briefly, each well of the 96-well microplate contained 190 − X μL of pH 7.0 buffer for thrombin (20 mM sodium phosphate containing 25 mM NaCl, 0.1 mM EDTA and 0.1% PEG8000) and pH 7.0 buffer for FXa (20 mM Tris-HCl, 2.5 mM $CaCl_2$, 100 mM NaCl, 0.1% PEG8000, 0.02% Tween 80) to which X μL of HS_{2S2S} (to give 0−25 μM final concentration) or reference was added followed by 5 μL of protease (to give 5 nM final concentration). After 10 min incubation at 25 °C, 5 μL of appropriate chromogenic substrate (to give 125 μM (Spectrozyme FXa) or 200 μM Spectrozyme TH) was rapidly added, and the residual protease activity was measured from the initial rate of increase in A_{405}. Relative residual protease activity at each concentration of HS_{2S2S} was calculated from the ratio of the activity in the presence and absence of the inhibitor. The dose−response profile was fitted by eq 3 to calculate the potency (IC_{50}) and efficacy ($\Delta Y = Y_M - Y_0$) of inhibition. In this equation, Y is the relative residual protease activity, Y_M and Y_0 are the maximum and minimum values of the relative residual protease activity, IC_{50} is the concentration of HS_{2S2S} that results in 50% inhibition of protease activity, and HC is the Hill slope.

$$Y = Y_0 + \frac{Y_m - Y_0}{1 + 10^{(\log[HS_{2S2S}]_0 - \log IC_{50})HC}} \quad (3)$$

■ ASSOCIATED CONTENT

Supporting Information

Details on computational studies, ESI-MS characterization, and structures of HS_{2S2S} tetrasaccharides. This material is available free of charge via the Internet at http://pubs.acs.org.

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: urdesai@vcu.edu.

Author Contributions

#R.S.B., A.S., and V.M.T. contributed equally to this work. R.S.B. performed thermodynamic and kinetic studies; A.S. and N.V.S. performed computational studies; V.M.T. synthesized HS_{2S2S} for the first time, which was reproduced and scaled up by T.K.N., M.Q., C.M., and A.J.; A.Y.M., and A.H.A. performed structural analysis; R.S.B., A.S., M.Q., A.Y.M., K.B., and U.R.D. contributed to manuscript preparation; U.R.D. compiled and finalized the paper.

Notes

The authors declare no competing financial interest.

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■ ABBREVIATIONS

AT, antithrombin; DEFGH, five residue heparin pentasaccharide sequence; FXa, factor Xa; GA, genetic algorithm; GAG, glycosaminoglycan; GlcA*p*2S, 2-O-sulfonated glucuronic acid; GLCE, C-5-epimerase; GlcN*p*, glucosamine residue; HCII, heparin cofactor II; HBS, heparin-binding site; HS, heparan sulfate; HS_{2S2S}, heparan sulfate containing sequences with GlcA*p*2S and GlcN*p*2S residues; HS2ST, 2-*O*-sulfotransferase; NDST, *N*-deacetylase/*N*-sulfotransferase; PAPS, phosphoadenosine phosphosulfate; RMSD, root-mean-square difference; TH, thrombin; UFH, unfractionated heparin

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